

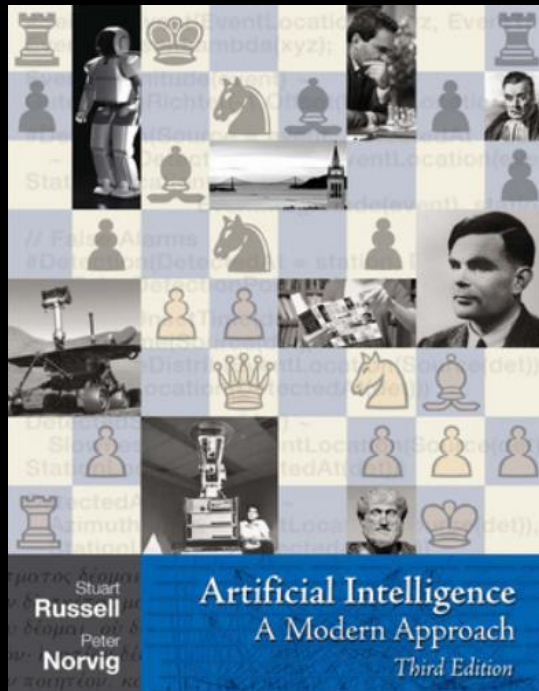
Artificial intelligence

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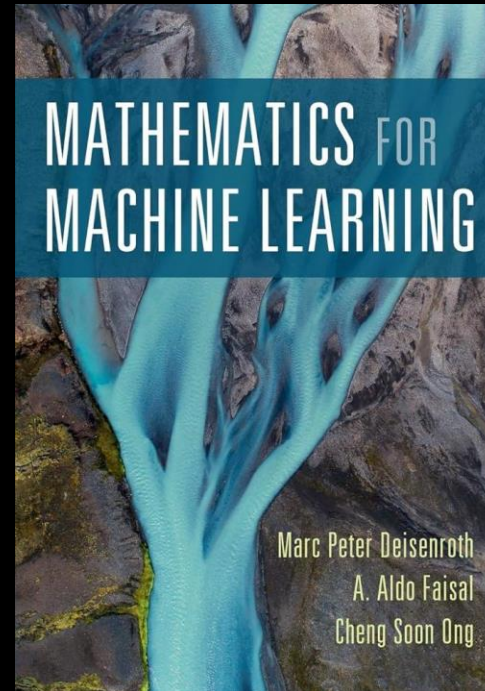


Course Overview

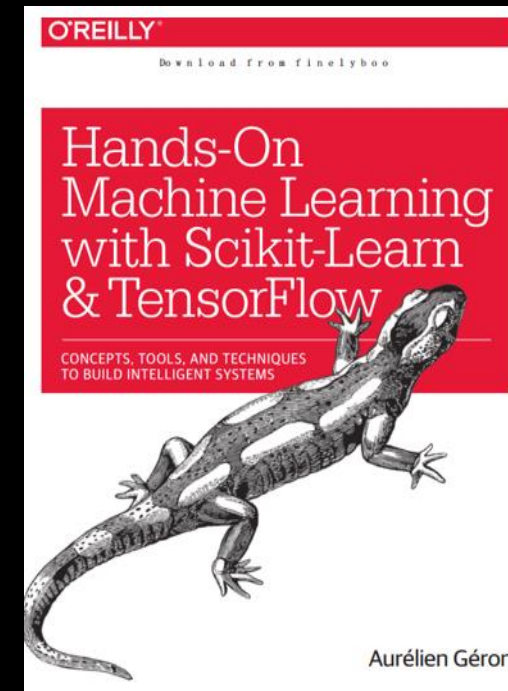
References



Artificial Intelligence



**Mathematics for
Machine Learning**



Machine learning

Course Overview

Class Progress and Evaluation

➤ Class progress

- PPT Lectures and discussion to achieve the objectives
- Attendance (in first 10 minutes)
- Exams
- Term Project (4~6 students in a team)

➤ Teaching Assistant and Office Hours

- Teaching Assistant: no now
- Office Hours: Monday & Wednesday (14:00 ~ 15:30)

Course Overview

Class Progress and Evaluation

➤ Evaluation

- Attendance: 10%
 - medical certificates are not accepted
 - 8 and more absences automatically **Fail**
- Mid-Term Exams: 25%
- Final Exam: 25%
- Homework-Assignment: 15%
- Term project: 25%

Course Overview

Term Project

➤ Evaluation

- Make a team (4~6 students in a team)
- End-to-end AI project
- Choose a project topic (free topic, should be not the same with other teams), (Week 5)
- Topics should be in these fields: Computer Vision, NLP, Generative AI, etc.
- Make a project proposal (Week 6)
- Prepare a report of the project
- Representation of the project (Week 14, might be in Week 16).

Course Overview

Syllabus

- **Week 1:** Introduction of Artificial Intelligence
- **Week 2:** Core ML Algorithms
- **Week 3:** ML for Classification
- **Week 4:** ML for Clustering
- **Week 5:** Introduction to Deep Learning
- **Week 6:** Convolutional Neural Networks
- **Week 7:** Advanced CV Models
- **Week 8:** **Mid-Term Exam**



Course Overview

Syllabus

- **Week 9:** Natural Language Processing
- **Week 10:** NLP with Transformers
- **Week 11:** Generative Models
- **Week 12:** Reinforcement Learning
- **Week 13:** Deep Reinforcement Learning
- **Week 14:** Term Project Presentation
- **Week 15:** **Final Exam**



Course Overview

About Me

Name: Oybek Eraliev Maripjon O'g'li

Bachelor Degree: Fergana Polytechnic Institute,
Electrical Engineering, 2008~2012.

Masters Degree: Fergana Polytechnic Institute,
Electrical Engineering, 2012~2014.

Ph.D degree: Inha University (South Korea),
Future Vehicle Engineering, 2019~2024.

Currently: Associate Professor in IUT, 2024~now



Course Overview

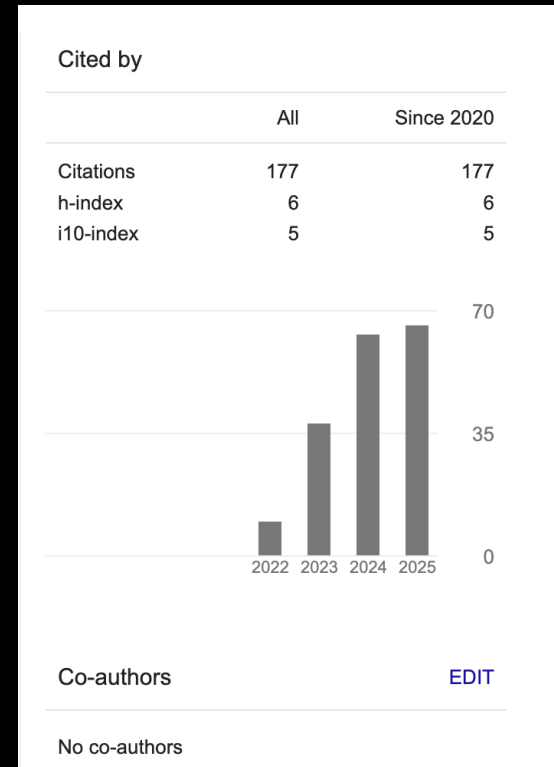
Researches



Oybek Eraliev 
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[Artificial Intelligent](#) [Machine/Deep Learning](#) [Diagnosis and Health monit...](#)



☐	TITLE	CITED BY	YEAR
☐	Sensing, perception, decision, planning and action of autonomous excavators OMU Eraliev, KH Lee, DY Shin, CH Lee Automation in Construction 141, 104428	76	2022
☐	Vibration-based loosening detection of a multi-bolt structure using machine learning algorithms O Eraliev, KH Lee, CH Lee Sensors 22 (3), 1210	38	2022
☐	Automatic optimization of one-dimensional CNN architecture for fault diagnosis of a hydraulic piston pump using genetic algorithm OEM Ugili, KH Lee, CH Lee IEEE Access 11, 68462-68472	22	2023
☐	Performance analysis of time series deep learning models for climate prediction in indoor hydroponic greenhouses at different time intervals O Eraliev, CH Lee Plants 12 (12), 2316	13	2023
☐	Experimental investigation on self-loosening of a bolted joint under cyclical temperature changes OMU Eraliev, YH Zhang, KH Lee, CH Lee Advances in Mechanical Engineering 13 (8), 16878140211039428	13	2021
☐	Self-loosening of a 3D-printed bolt by using three different materials under cyclical temperature changes O Eraliev, KH Lee, CH Lee Applied Sciences 12 (6), 3001	6	2022
☐	SCAE—Stacked convolutional autoencoder for fault diagnosis of a hydraulic piston pump with limited data samples O Eraliev, KH Lee, CH Lee Sensors 24 (14), 4661	5	2024
☐	Reliable fault diagnosis method based on an optimized deep belief network for gearbox O Eraliev, O Xakimov, CH Lee 드라이브·컨트롤 20 (4), 54-63	4	2023



Research Interests:
Applied AI, Machine/Deep Learning, NLP.



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Course Overview

Researches



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Introduction of Artificial Intelligence

Part – 1

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Content

- What Is AI?
- The Foundations of Artificial Intelligence
- The State of the Art
- Risks and Benefits of AI

What Is AI?

Artificial

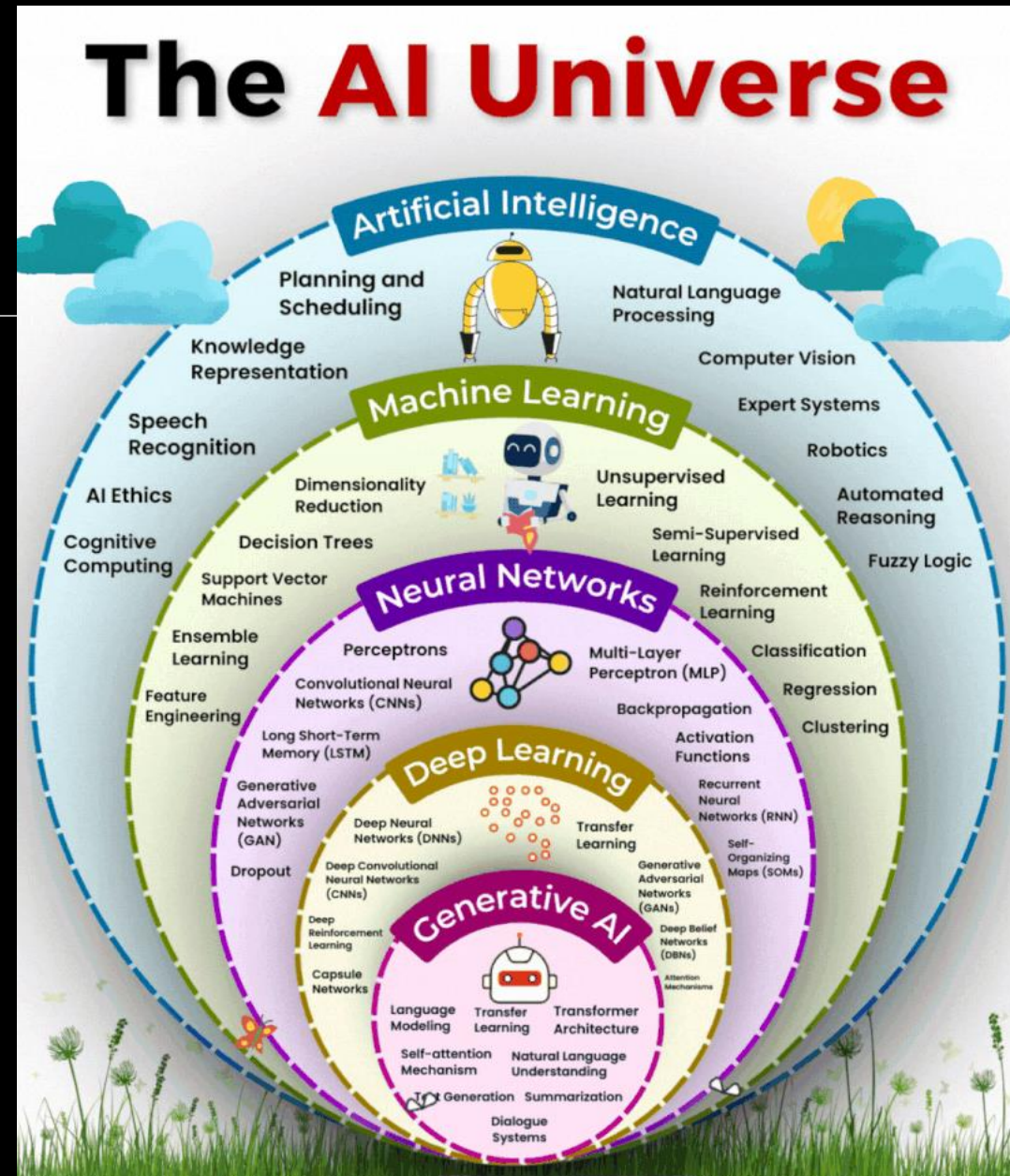
made or produced by human beings rather than occurring naturally, especially as a copy of something natural.

What Is AI?

Intelligence

the ability to acquire and apply knowledge and skills.

What Is AI?



What Is AI?

- ✍ Some have defined intelligence in terms of fidelity to *human* performance, while others prefer an abstract, formal definition of intelligence called *rationality* (doing the “right thing”)
- ✍ Some consider intelligence to be a property of internal *thought* processes and reasoning, while others focus on intelligent *behavior*, an external characterization.

What Is AI?

✍ From these two dimensions—*human* vs. *rational* and *thought* vs. *behavior*—there are four possible combinations, and there have been adherents and research programs for all four.

① Thinking humanly

③ Thinking rationally

② Acting humanly

④ Acting rationally

What Is AI?

Thinking humanly

✍️ ***The Turing test***, proposed by Alan Turing (1950), was designed as a thought experiment that would sidestep the philosophical vagueness of the question “***Can a machine think?***”

✍️ A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.

What Is AI?

Thinking humanly

✍️ Programming a computer to pass an applied test provides plenty to work on. The computer would need the following *capabilities*:

- ✍️ **Natural language processing** to communicate successfully in a human language;
- ✍️ **Knowledge representation** to store what it knows or hears;
- ✍️ **Automated reasoning** to answer questions and to draw new conclusions;
- ✍️ **Machine learning** to adapt to new circumstances and to detect and extrapolate patterns.

What Is AI?

Thinking humanly

✍ However, other researchers have proposed a **total Turing test**, which requires interaction with objects and people in the real world. To pass the total Turing test, a robot will need :

✍ **Computer vision and speech recognition** to perceive the world;

✍ **Robotics** to manipulate objects and move about.

What Is AI?

Thinking humanly

➤ These six disciplines compose **most of AI**.

- 1 Natural language processing
- 2 Knowledge representation
- 3 Automated reasoning
- 4 Machine learning
- 5 Computer vision and speech recognition
- 6 Robotics

Content

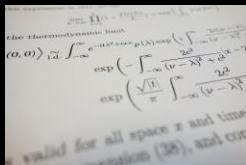
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The Foundations of Artificial Intelligence

Philosophy



Mathematics



Economics



Neuroscience



Psychology



Computer engineering



Control theory and cybernetics



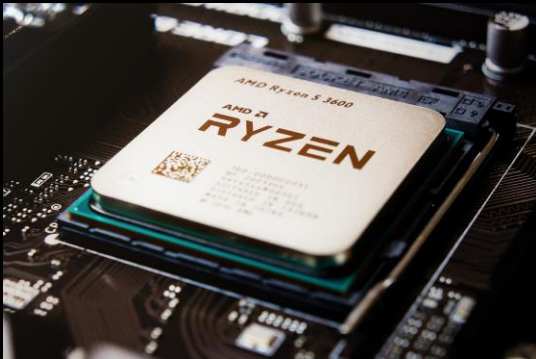
Linguistics



The Foundations of Artificial Intelligence

Computer engineering

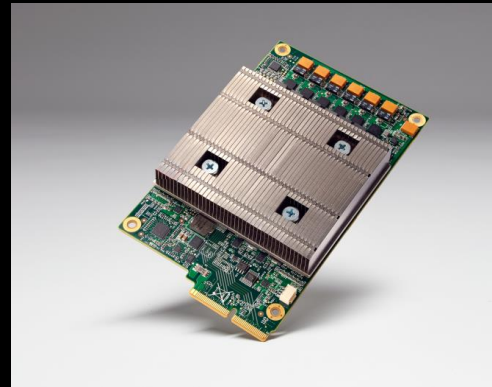
How can we build an efficient computer?



CPU



GPU



TPU



Quantum Computing

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The State of the Art

Publications

Human benchmarks

Students

Industry

Sentiment

Vision

Diversity

Speed

Conferences

Language



The State of the Art

What can AI do today?

Robotic vehicles:

In 2018, Waymo test vehicles passed the landmark of 10 million miles driven on public roads without a serious accident, with the human driver stepping in to take over control only once every 6,000 miles.

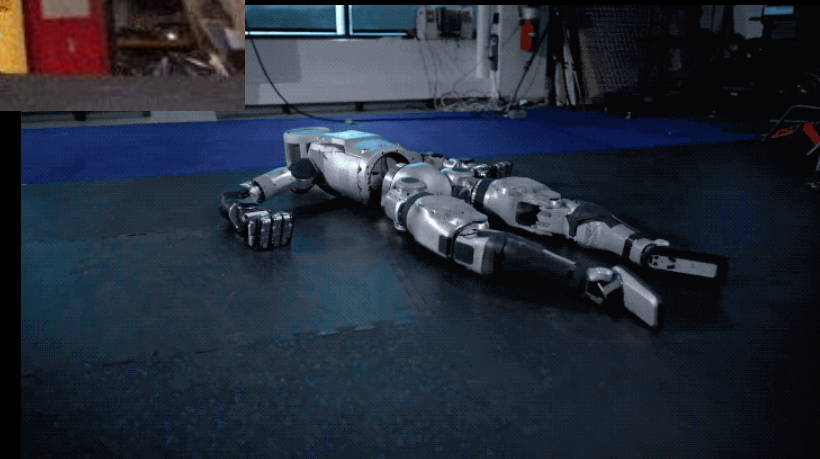
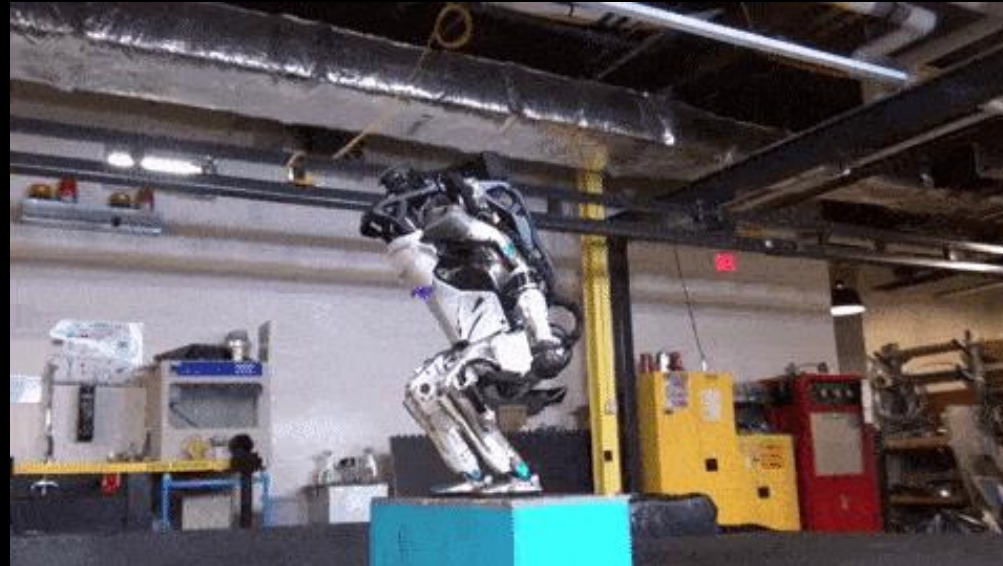


The State of the Art

What can AI do today?

Legged locomotion:

Atlas, a humanoid robot, not only walks on uneven terrain but jumps onto boxes and does backflips.

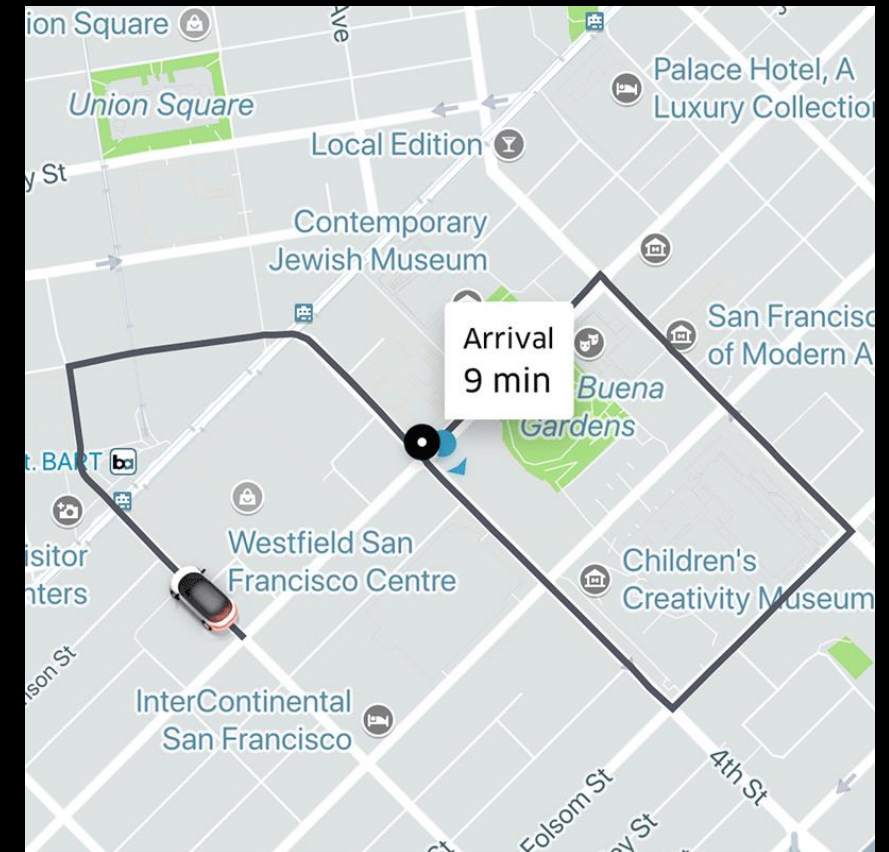


The State of the Art

What can AI do today?

Autonomous planning and scheduling:

Every day, ride hailing companies such as Uber, Yandex and mapping services such as Google Maps provide driving directions for hundreds of millions of users, quickly plotting an optimal route taking into account current and predicted future traffic conditions.

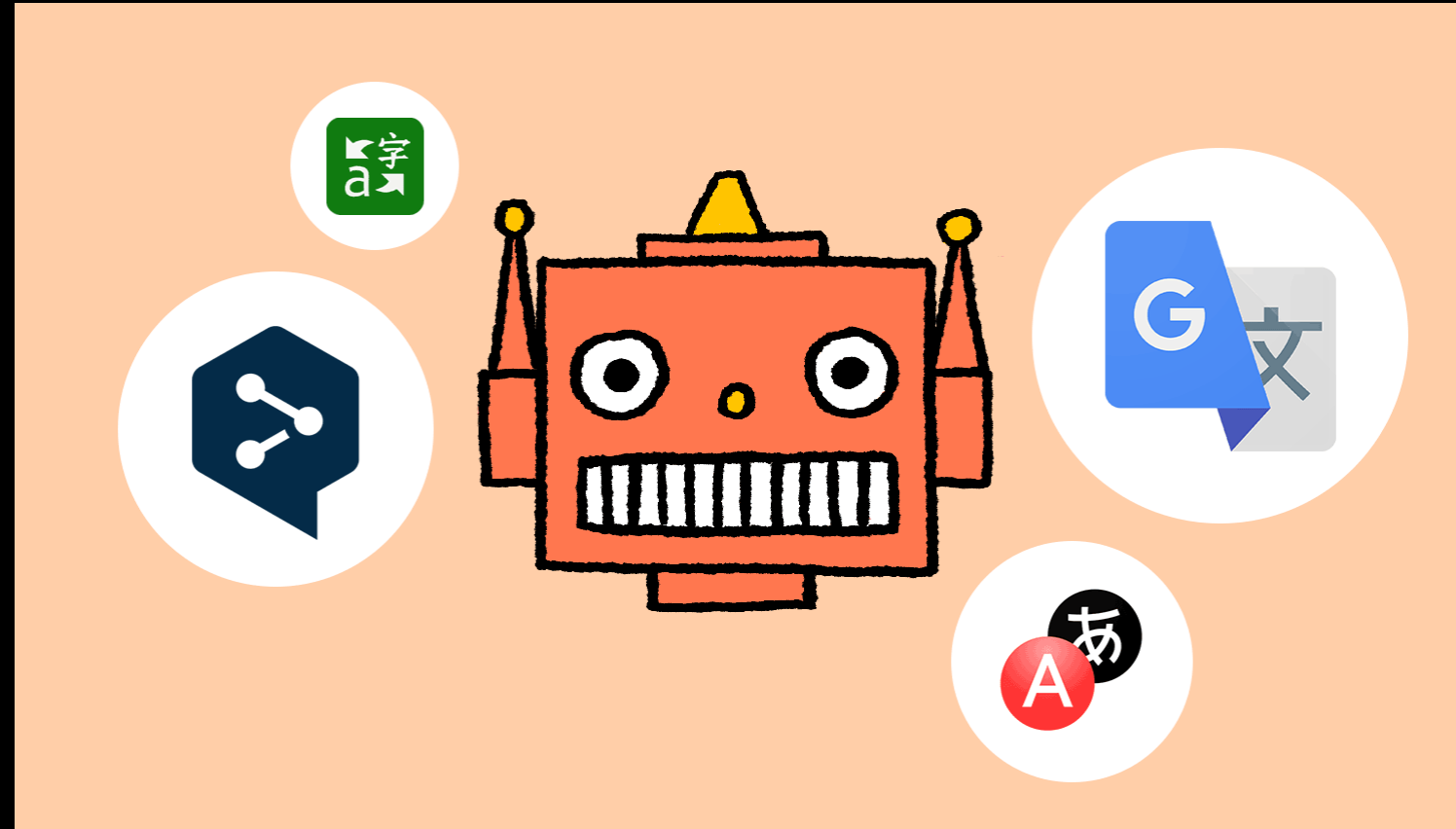


The State of the Art

What can AI do today?

Machine translation:

Online machine translation systems now enable the reading of documents in over 100 languages, including the native languages of over 99% of humans, and render hundreds of billions of words per day for hundreds of millions of users.



The State of the Art

What can AI do today?

Speech recognition:

Alexa, Siri, Cortana, and Google offer assistants that can answer questions and carry out tasks for the user.

For example, the Google Duplex service uses speech recognition and speech synthesis to make restaurant reservations for users, carrying out a fluent conversation on their behalf.

ALEXA vs SIRI vs GOOGLE

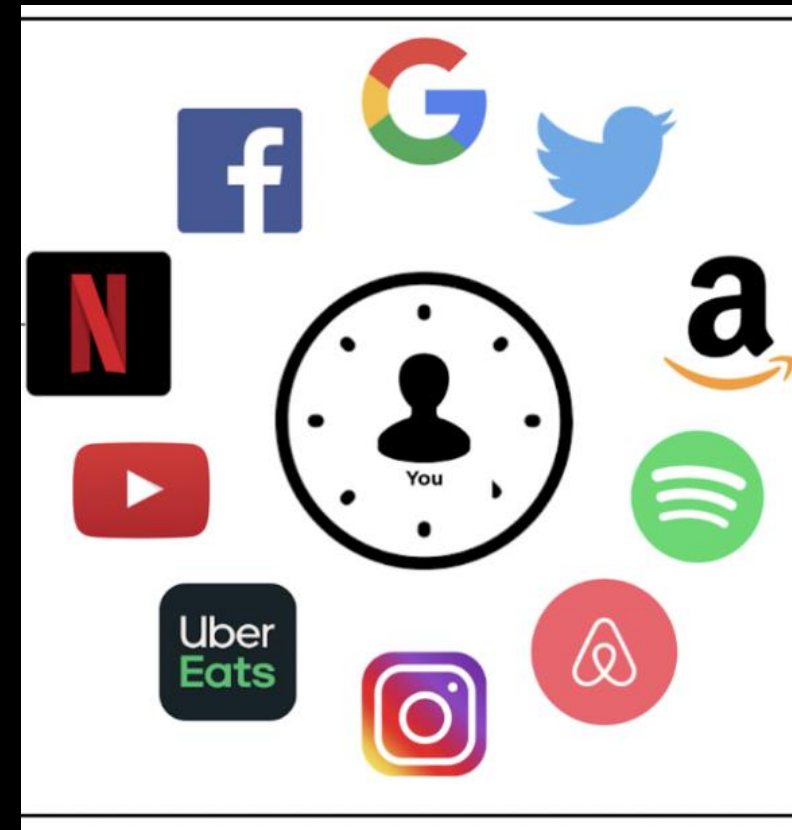


The State of the Art

What can AI do today?

Recommendations:

Companies such as Amazon, Facebook, Netflix, Spotify, YouTube, Walmart, and others use machine learning to recommend what you might like based on your past experiences and those of others like you.



The State of the Art

What can AI do today?

Medicine:

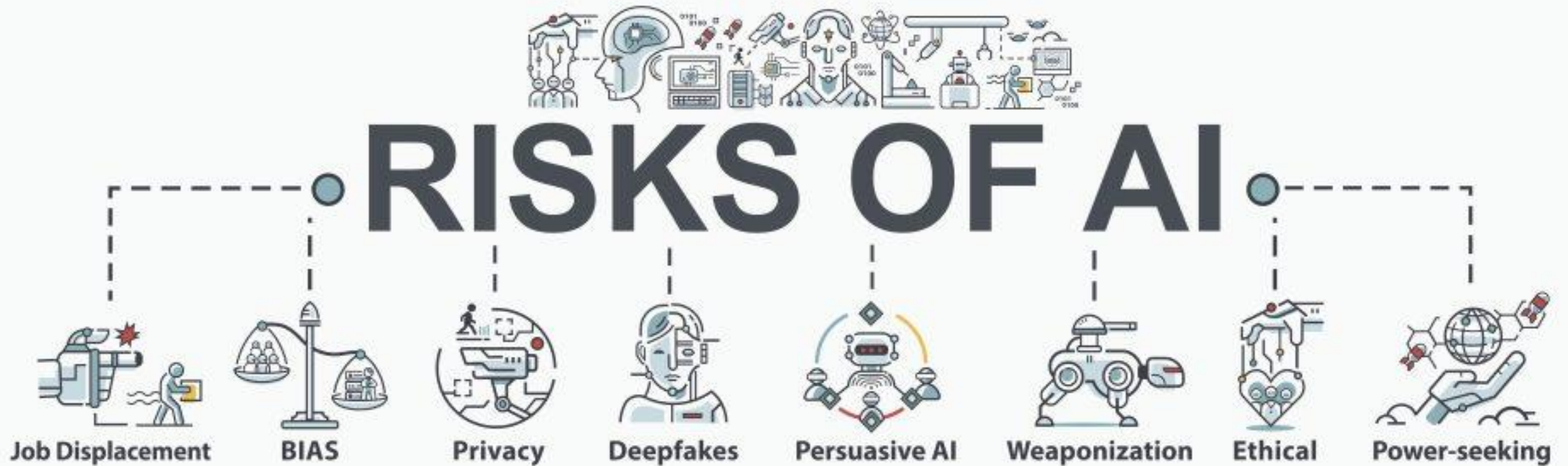
AI algorithms now equal or exceed expert doctors at diagnosing many conditions, particularly when the diagnosis is based on images. Examples include Alzheimer's disease (Ding *et al.*, 2018), metastatic cancer (Liu *et al.*, 2017; Esteva *et al.*, 2017), ophthalmic disease (Gulshan *et al.*, 2016), and skin diseases (Liu *et al.*, 2019c).



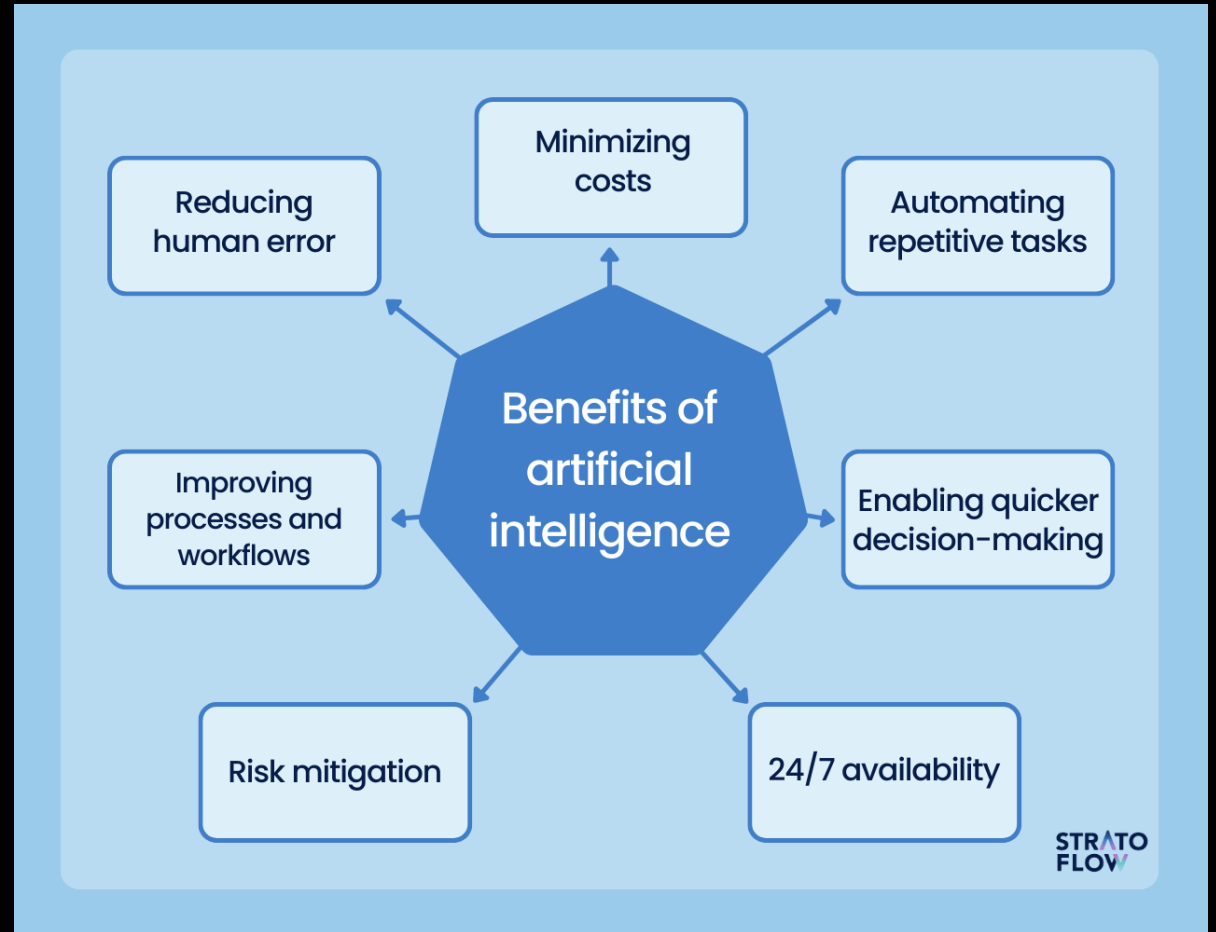
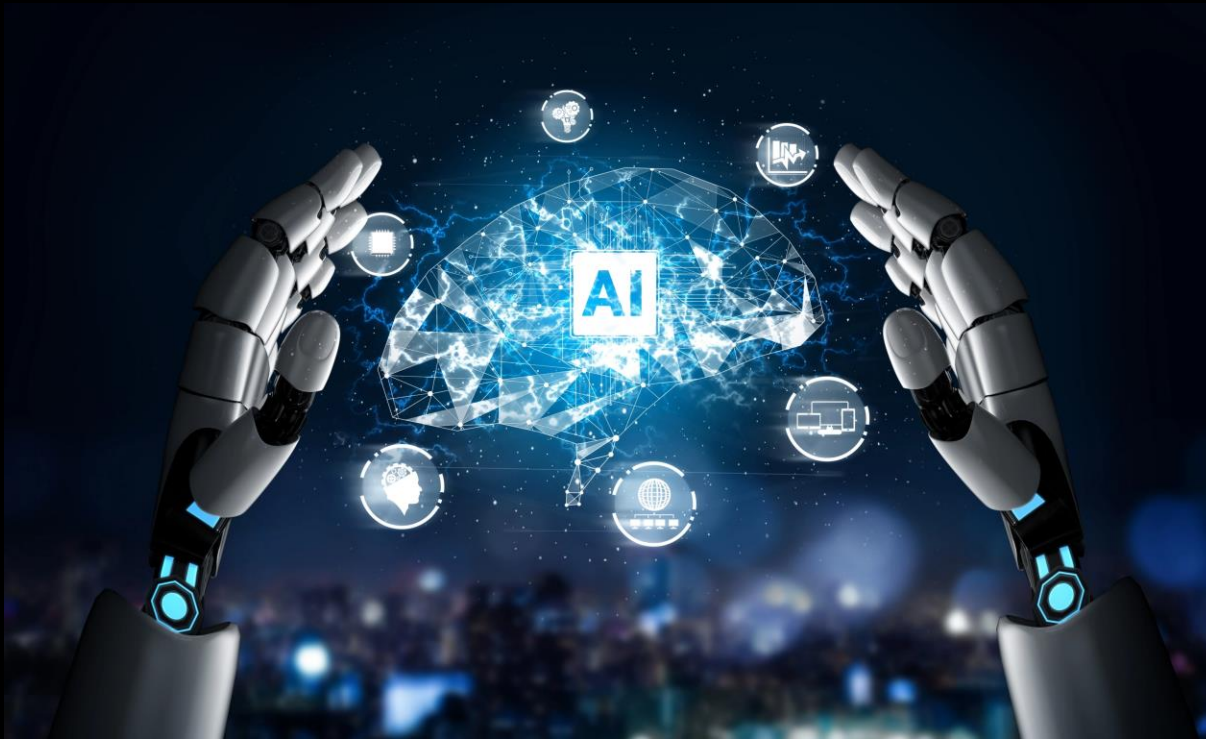
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Risks and Benefits of AI



Risks and Benefits of AI



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